Lecture 19 (10/30)

Interactions and residual plots

The course so far

So far:

- **Unit 1:** terminology and notation (probability, random variables, conditioning).
- **Unit 2:** linear regression as a model of conditional probabilities,

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

- **Unit 3:** how to fit the model and quantify uncertainty (estimators, standard errors, tests and confidence intervals).
- **Unit 4:** Multiple predictor linear regression,

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  and applications.

Next:

- **Unit 5:** nonlinear models and overfitting. Machine learning vs. statistics.
- **Unit 6:** Bayesian statistics.

Today:

1) Interactions (test scores, mother's schooling and IQ)

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- The effect of a drug for an infection might depend on whether someone has had the infection before.

These are interactions and are not captured by our existing models.

Example

Test scores, revisited. Survey of 434 children (from *Regression and Other Stories*):

- Y (`kid_score`): the child's test score;
- X_{hs} (`mom_hs`): 1 if the mother attended high school, 0 if not;
- X_{iq} (`mom_iq`): the mother's IQ.

Without interactions, the model output was

```
Dep. Variable:              kid_score   R-squared:                      0.214
                  coef    std err          t      P>|t|      [0.025      0.975]
const          25.7315      5.875      4.380      0.000      14.184      37.279
mom_hs          5.9501      2.212      2.690      0.007       1.603      10.297
mom_iq          0.5639      0.061      9.309      0.000       0.445       0.683
```

Interaction terms

Let the slope vs. X_{iq} depend linearly on X_{hs}:

$$\begin{aligned} Y &= \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + (\beta_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}}) X_{\text{iq}} + \epsilon \\ &= \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} X_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}} X_{\text{iq}} + \epsilon. \end{aligned}$$

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Question

Compute

$$E[Y \mid X_{\text{hs}}{=}1, X_{\text{iq}} = x] - E[Y \mid X_{\text{hs}}{=}0, X_{\text{iq}} = x].$$

Fitting the interaction model

Dep. Variable:		kid_score	R-squared:			0.230
	coef	std err	t	P>|t|	[0.025	0.975]
const	-11.4820	13.758	-0.835	0.404	-38.523	15.559
mom_hs	51.2682	15.338	3.343	0.001	21.122	81.414
mom_iq	0.9689	0.148	6.531	0.000	0.677	1.260
hs_iq	-0.4843	0.162	-2.985	0.003	-0.803	-0.165

Here `hs_iq` is the new predictor $X_{\text{hs}}X_{\text{iq}}$.

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Question

- What is the slope of score vs. mother's IQ in each schooling group?
- How does the effect of high school depend on IQ?
- Why is $\hat{\beta}_{\text{hs}} \approx 51$ so different from $\hat{\beta}_{\text{hs}} \approx 5.95$ in the model without the interaction?

Centering a predictor

Replace X_{iq} by the centered predictor $\Delta_{\text{iq}} = X_{\text{iq}} - \overline{X}_{\text{iq}}$ (here $\overline{X}_{\text{iq}} = 100$) and fit

$$Y = \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} \Delta_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}} \Delta_{\text{iq}} + \epsilon.$$

Dep. Variable:		kid_score	R-squared:			0.230
	coef	std err	t	P>|t|	[0.025	0.975]
const	85.4069	2.218	38.502	0.000	81.047	89.767
mom_hs	2.8408	2.427	1.171	0.242	-1.929	7.610
iq_c	0.9689	0.148	6.531	0.000	0.677	1.260
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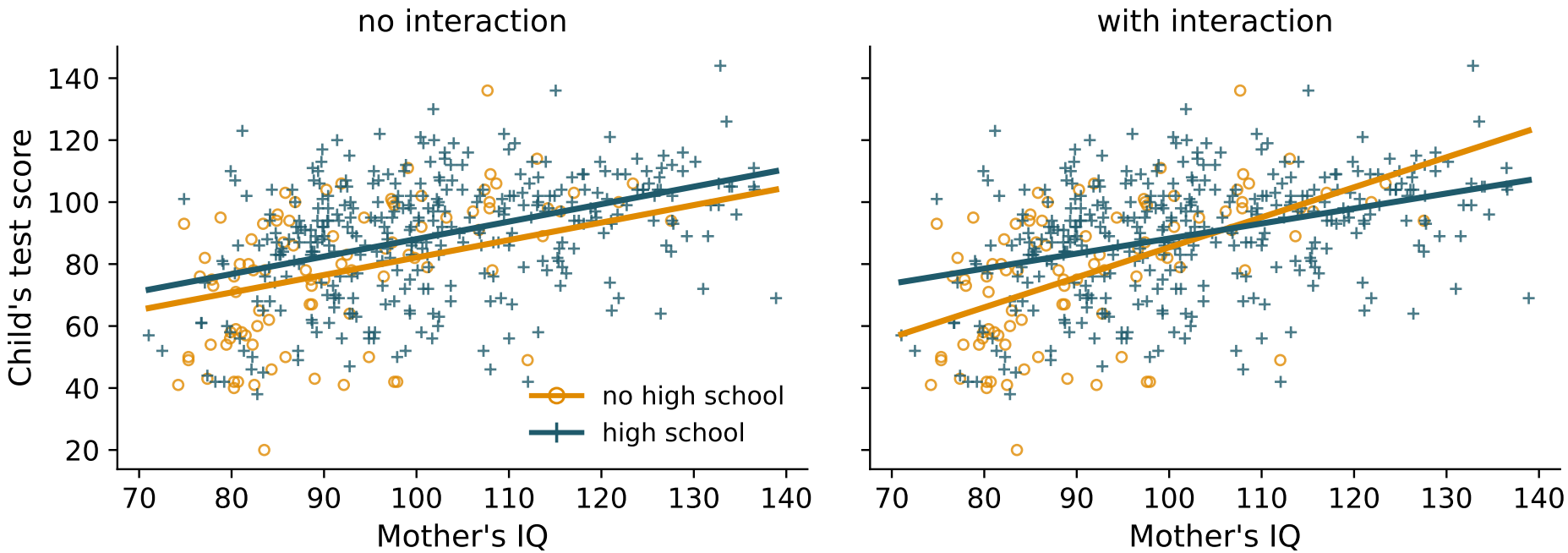
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Question

Interpret $\hat{\beta}_0$ and $\hat{\beta}_{\text{hs}}$ in the output above.

Visualizing regression lines with interaction



Without the interaction the fitted lines are parallel.

Example: earnings, height and gender

Survey of 1629 adults with positive earnings (from *Regression and Other Stories*): Y is the log of earnings in \$1000s, X_{height} is height in inches, and $X_{\text{male}} = 1$ for men, 0 for women. Without an interaction:

Dep. Variable:		log_earnings	R-squared:			0.080
	coef	std err	t	P>|t|	[0.025	0.975]
const	1.0686	0.506	2.111	0.035	0.076	2.062
height	0.0238	0.008	3.044	0.002	0.008	0.039
male	0.3715	0.062	6.023	0.000	0.251	0.493

With an interaction, and height centered at its mean (66.8 inches):

Dep. Variable:		log_earnings	R-squared:			0.081
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.6405	0.037	71.856	0.000	2.568	2.713
height_c	0.0148	0.011	1.389	0.165	-0.006	0.036
male	0.3568	0.063	5.684	0.000	0.234	0.480
male_height_c	0.0196	0.016	1.251	0.211	-0.011	0.050

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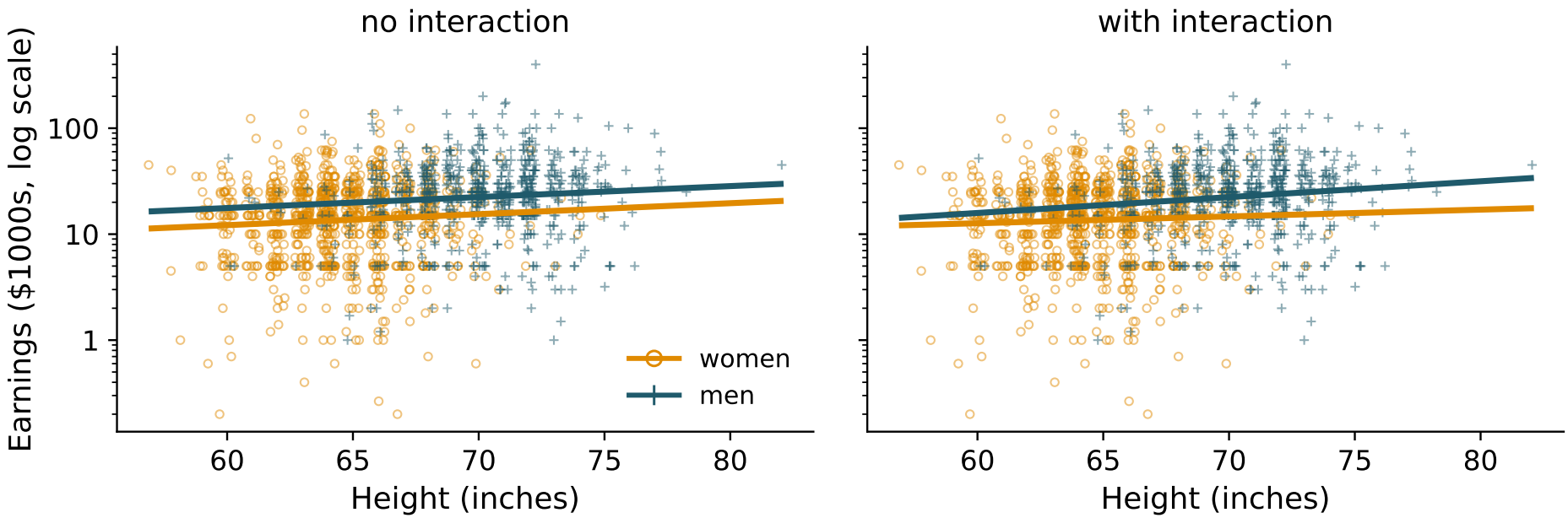
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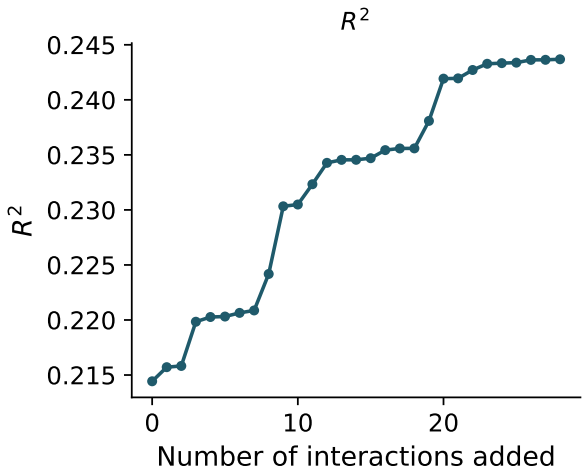
What is the slope of log earnings vs. height for women? For men? What is the difference between men and women of average height? Interpret each as a percent change in earnings.

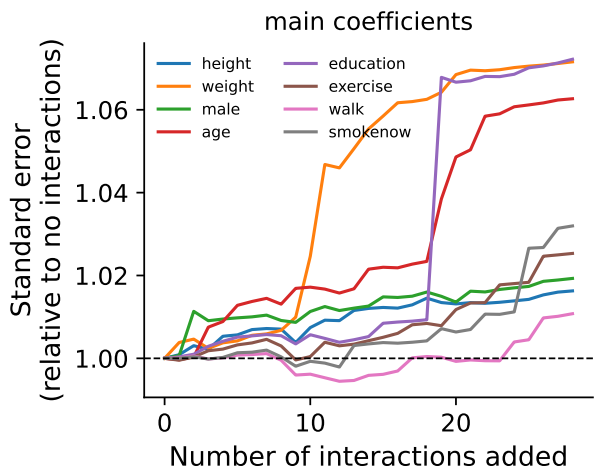
Earnings: no interaction vs. interaction

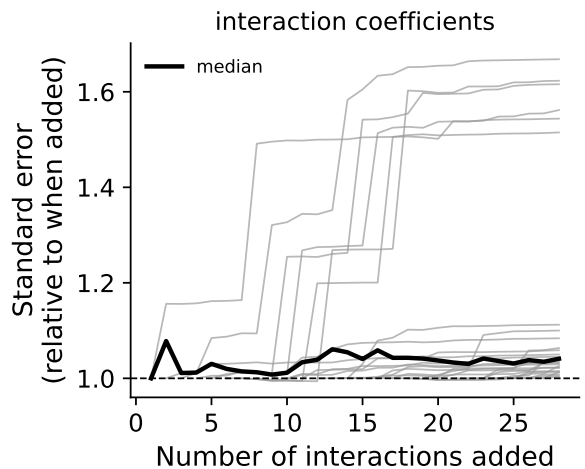


Earnings: adding every interaction

Consider log earnings with eight predictors: height, weight, gender, age, education, exercise, walking and smoking. There are 28 possible pairwise interactions. Below, they are added one at a time, using all 1604 people with complete data.

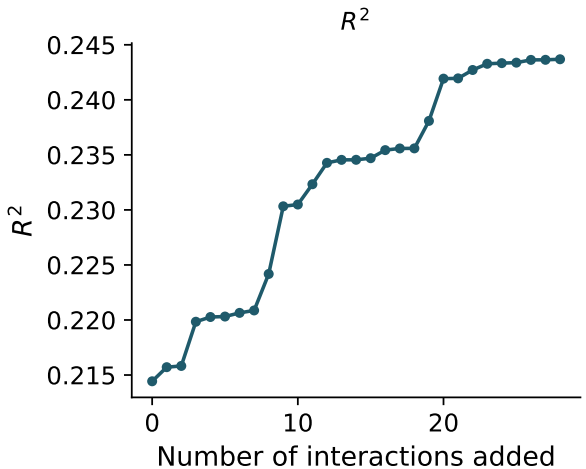


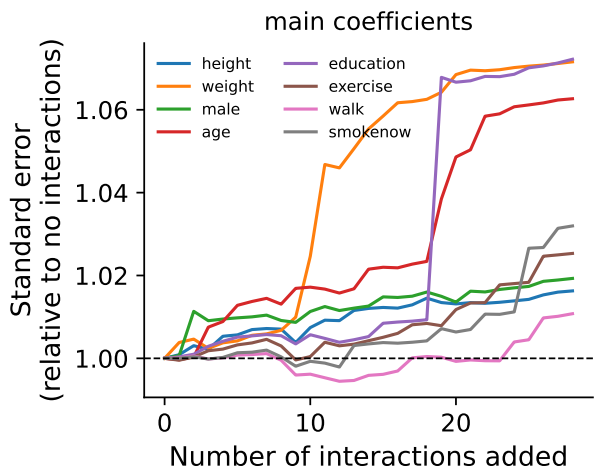


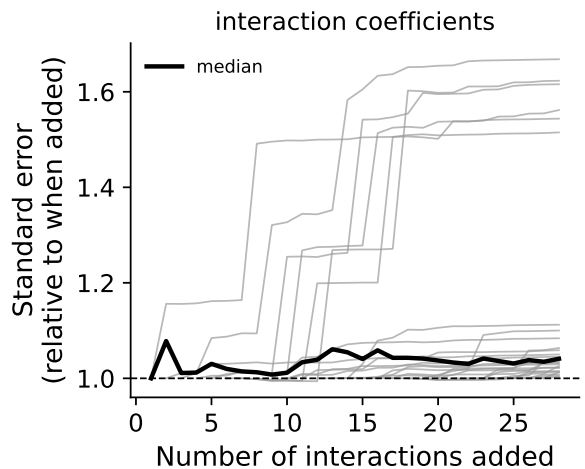


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Question

Is the model with all 28 interactions better than the one with none? What happens to the uncertainty in each coefficient?